CO2 – ASSESSMENT 3

SCENARIO-BASED ASSESSMENT

Course: Ethical Hacking **Course Code:** ITA1406
Slot: Slot B **Total Marks:** 100
Course Faculty: Dr. K. Anbazhagan
Topic: DNS Footprinting using DNSDumpster

1. Objective

To perform DNS footprinting for the assigned organization using DNSDumpster, review publicly visible DNS and host information, interpret the generated infrastructure map, and submit the downloaded Excel report and map. The handout asks students to find their assigned website from the register-number table, run a test, download the report as .xlsx, and upload the map.

2. Assigned domain

The assignment lists 40 possible websites mapped to register numbers on pages 1–2. The student's register number was not provided with the uploaded handout, so the assigned domain cannot be determined accurately. The report therefore does not guess a company or fabricate DNS records, hostnames, IP addresses, or an Excel export.

3. Procedure

1. Open DNSDumpster: https://dnsdumpster.com
2. In “Enter a Domain to Test”, type the website assigned to your register number.
3. Click “Start Test”.
4. Review the DNS records and host information displayed.
5. Download the complete report as an .xlsx file.
6. Save the generated DNS map and include it with the report.

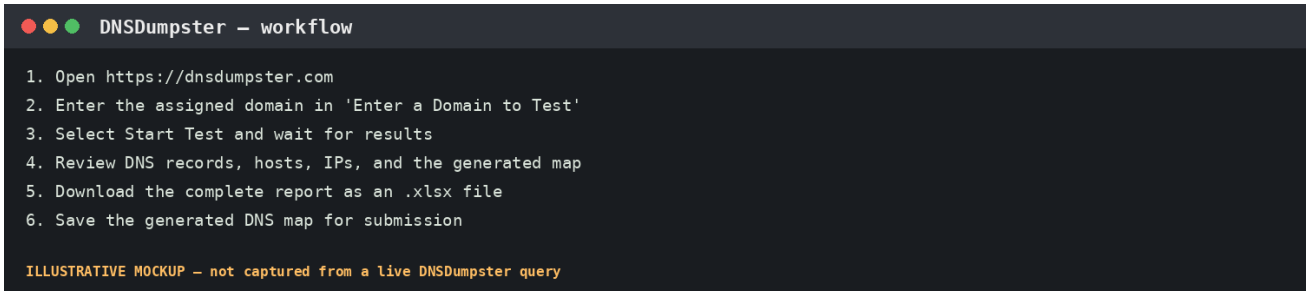


Figure 1. Illustrative workflow mockup, not a real DNSDumpster result screenshot.

4. DNS information and interpretation

Record / field	Meaning	How to report it
A record	Maps a hostname to an IPv4 address.	Copy hostname and IPv4 address exactly as displayed.
AAAA record	Maps a hostname to an IPv6 address.	Record the hostname and IPv6 address if present.
MX record	Identifies mail-exchange servers for a domain.	List the mail server hostnames shown.
NS record	Identifies authoritative name servers.	List the name server hostnames shown.
CNAME record	Aliases one hostname to another hostname.	Record the alias and canonical target if displayed.
Host / IP	A discovered hostname and its associated address.	Use the DNSDumpster output; do not estimate values.

5. DNS map screenshot / interpretation

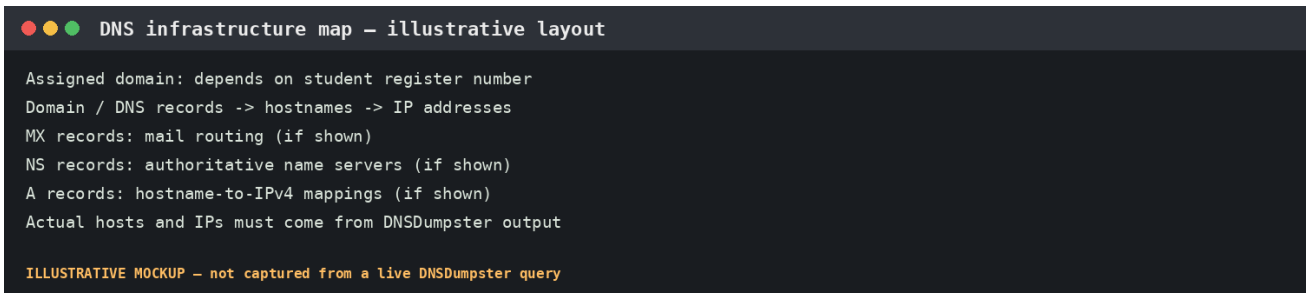


Figure 2. Illustrative DNS map concept. It is not a map generated by DNSDumpster and does not represent actual infrastructure.

A DNS map helps visualize the relationship between a domain, its discovered hostnames, and associated IP addresses. Mail-exchange (MX) records can indicate mail routing, while name-server (NS) records identify authoritative DNS providers. A and AAAA records show address mappings; CNAME records indicate aliases. The actual map may differ depending on the assigned domain and the records visible at the time of testing.

6. Security analysis

DNS footprinting can reveal public-facing services and infrastructure relationships. This information can help an organization maintain an accurate asset inventory, but an attacker could also use it to select targets for phishing, service enumeration, or further reconnaissance. DNS results alone do not prove that a system is vulnerable.

Recommended precautions

- Review public DNS records regularly and remove obsolete records.
- Avoid publishing unnecessary internal hostnames or sensitive infrastructure details.
- Keep exposed services patched and restrict administrative access.
- Monitor DNS changes and investigate unexpected records.
- Use this information only for authorized educational assessment and defensive review.

7. Result

The DNSDumpster procedure and the interpretation of common DNS record types are documented. Actual hostnames, IP addresses, DNS map, and .xlsx report are not included because the assigned domain is unknown and no live DNSDumpster test was performed.

8. Conclusion

DNSDumpster can help collect publicly visible DNS and host information and present relationships in a map. Correct analysis requires using the domain assigned to the student's register number, downloading the actual report, and interpreting only verified records.

9. Submission checklist

- Confirm the assigned domain using the register-number table.
- Run the DNSDumpster test.
- Download the complete .xlsx report.
- Save the DNS map image.
- Review host/IP and DNS records and include the actual evidence.

10. Reference

- DNSDumpster: https://dnsdumpster.com
- Original handout: CO2 – Assessment 3, Scenario-Based Assessment, Ethical Hacking, ITA1406, Slot B.

The original rubrics page from the supplied assignment is appended at the end.

RUBRICS

Criteria	Weightage (%)	Level 4 (Exemplary)	Level 3 (ProVicient)	Level 2 (Developing)	Level 1 (Beginning)
DNS Footprinting Execution	25%	Successfully performs DNS footprinting on the assigned domain using DNSDumpster and follows the complete procedure accurately.	Performs DNS footprinting correctly with minor procedural errors.	Performs basic DNS footprinting with assistance.	Unable to perform the required DNS footprinting.
DNS Information IdentiVication	25%	Accurately identiQies and interprets relevant DNS, host, IP, and infrastructure information from the generated results.	IdentiQies most relevant information with minor omissions.	IdentiQies some DNS information but provides limited interpretation.	Fails to identify or interpret relevant DNS information.
Analysis of DNS Infrastructure	30%	Thoroughly analyzes the DNS map and discovered infrastructure and explains its security signiQicance clearly.	Correctly analyzes most infrastructure information with minor gaps.	Provides basic analysis but misses important relationships or security implications.	Unable to analyze the DNS infrastructure meaningfully.
Report and Evidence Presentation	20%	Submits a complete Excel report and DNS map with clear, accurate, and well-organized evidence.	Provides the required report and map with minor documentation issues.	Submission is partially complete or poorly organized.	Required report/map is missing, incomplete, or unusable.